HEALTHCARE TECHNOLOGY | SOCIAL IMPACT

Blood Donor Finder

A MERN Stack Web Application

A web application that helps users discover potential blood donors based on blood group and location.

Connecting Donors. Saving Lives.



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The Challenge

Emergencies are urgent

Finding suitable donors quickly can be difficult.

Information is scattered

Donor details sit across different sources.

Manual search is slow

Searching by hand wastes valuable time.

Key challenges

- Identifying potential matches
- Maintaining accurate records
- Protecting contact information
- Keeping availability updated

The need: a centralized, easy-to-use donor search platform

Our Innovation

Donor Registration

Name, blood group, location and contact details.

Search & Filter

Find donors by blood group and location.

View Matches

Browse matching donor records.

CRUD Operations

Add, view, update, delete records (authorized).

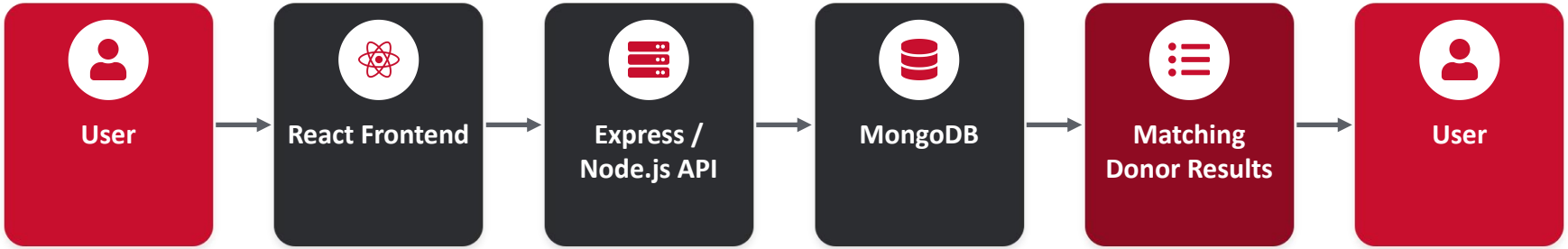
Secure Access

Authentication and role-based access where required.

Our Approach

Donor management and targeted search in one responsive web app.

How It Works



1

Donor registers with the required information

2

User selects a blood group and optionally a location

3

Frontend sends a search request to the API

4

Backend queries MongoDB for matching records

5

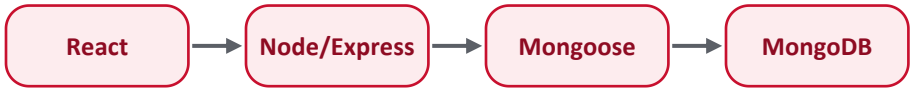
Results shown per access and privacy rules

User journey: Register → Search → View matching donors

Technology & Algorithm

MongoDB	Stores donor and user records
Express.js	Handles API routes and requests
React.js	Builds the interactive UI
Node.js	Runs the backend server
Mongoose	Schemas and MongoDB access
JWT / sessions	Authentication, where implemented

ARCHITECTURE



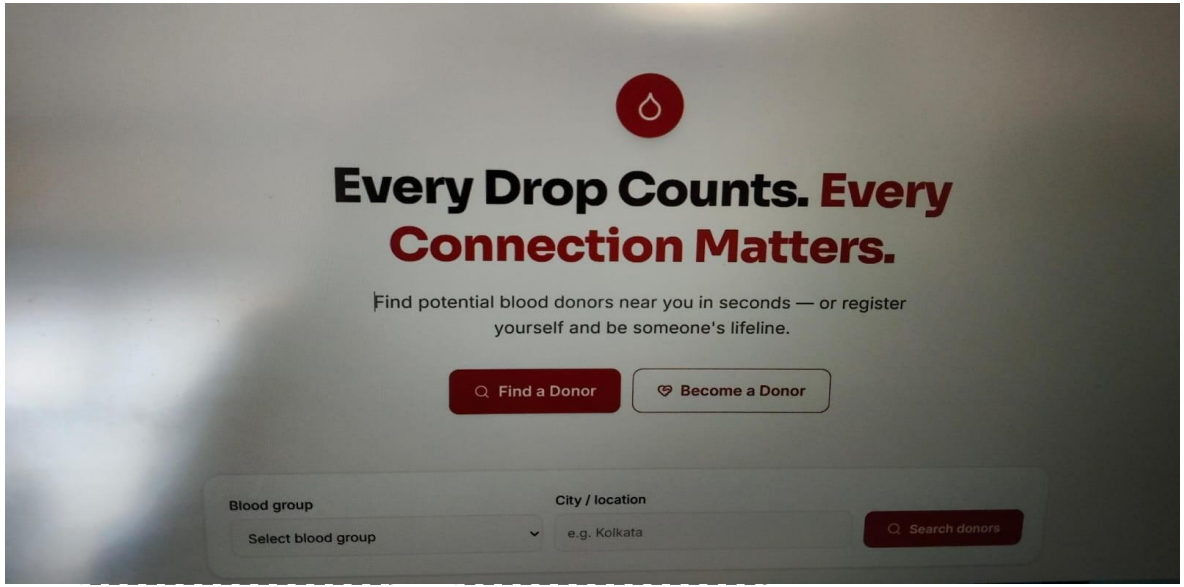
Search algorithm

1. Accept blood group + optional location
2. Validate and normalize the inputs
3. Query: exact blood-group match + location filter
4. Return a structured response

Complexity

- With a suitable index: efficient (depends on query, index and data)
- Full collection scan: O(n)
- Optional sort: O(k log k) for k matches

Demo & Impact



SAMPLE OUTPUT (ILLUSTRATIVE)

Search

Blood Group = O+
Location = Hyderabad

Result: matching registered donors

- Donor A · O+ · Hyderabad
- Donor B · O+ · Hyderabad
- Donor C · O+ · Hyderabad

- Easier discovery of potential donors
- Centralized donor information
- Less manual searching effort
- Simple register and search interface

Future Vision

Current achievement

- Donor registration
- Blood group + location search
- CRUD operations
- Database connectivity

[Edit to match your prototype]

Future improvements

- OTP / email verification
- Maps and location search
- Availability status + timestamps
- Consent-based emergency alerts
- Mobile app integration
- Stronger privacy controls

Scalability: indexes, pagination, API security, monitoring, cloud hosting

“A simple technology solution to help connect people with potential blood donors.”